



2026 年 01 月 CSCA 物理考试

January 2026 CSCA Physics Exam

题目(Questions)、答案(Answers)与解析(Explanations)

第 1 题(Question 1)

题目:第 26 届国际计量大会决定重新定义质量的基本单位, 该定义于 2019 年 5 月 20 日生效。

质量的基本单位是(The 26th International Conference on Weights and Measures decided to redefine the basic unit of mass, which came into effect on May 20, 2019. The basic unit of mass is)

- A. N(牛顿, newton)
- B. s(秒, second)
- C. m(米, meter)
- D. kg(千克, kilogram)

第 2 题(Question 2)

题目:月球表面的重力加速度约为地球表面的 $\frac{1}{6}$ 。若某物体在地球上的重力为 120 牛, 则其在月球上的重力为(The gravitational acceleration on the surface of the moon is about $\frac{1}{6}$ of that on Earth. If the gravity of an object on Earth is 120N, its gravity on Moon is)

- A. 60N(60 牛, sixty newtons)
- B. 20N(20 牛, twenty newtons)
- C. 120N(120 牛, one hundred and twenty newtons)
- D. 720N(720 牛, seven hundred and twenty newtons)

第 3 题(Question 3)

题目:一根弹簧的原长为 0.7 米, 劲度系数 $k = 10, \text{N/m}$ 。当施加 3 牛的力拉伸该弹簧时, 其新长度为(The natural length of a spring is 0.7m, and its spring constant is $k = 10, \text{N/m}$. When a force of 3N is applied to stretch the spring, its new length is)

- A. 0.5m(0.5 米, 0.5 meters)
 - B. 0.7m(0.7 米, 0.7 meters)
 - C. 0.3m(0.3 米, 0.3 meters)
 - D. 1.0m(1.0 米, 1.0 meters)
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第 4 题(Question 4)

题目:假设某物体受到两个力 F_1 和 F_2 的作用, 已知 $F_1 = 3, \text{N}$ 、 $F_2 = 5, \text{N}$, 则这两个力的合力大小不可能为(Suppose that an object is subjected to two forces F_1 and F_2 . Given that $F_1 = 3, \text{N}$ and $F_2 = 5, \text{N}$, the magnitude of the resultant force of these two forces cannot be)

- A. 2N(2 牛, two newtons)
 - B. 9N(9 牛, nine newtons)
 - C. 5N(5 牛, five newtons)
 - D. 8N(8 牛, eight newtons)
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第 5 题(Question 5)

题目:在匀强磁场中, 一根长度为 0.5 米的直导线垂直于磁场方向放置, 当通过 10 安的电流时, 导线受到 1.5 牛的安培力。该磁场的磁感应强度大小为(In a uniform magnetic field, a straight wire with length 0.5m is placed perpendicular to the direction of the magnetic field. When passing through a current of 10A it experiences an Ampere force of 1.5N. The magnitude of the



magnetic induction intensity of the magnetic field is)

- A. 0.3T(0.3 特斯拉, 0.3 tesla)
 - B. 0.2T(0.2 特斯拉, 0.2 tesla)
 - C. 0.4T(0.4 特斯拉, 0.4 tesla)
 - D. 0.1T(0.1 特斯拉, 0.1 tesla)
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第 6 题(Question 6)

题目:乒乓球非常适合亚洲人的身体特点, 是中国的国球。下列关于乒乓球的说法正确的是

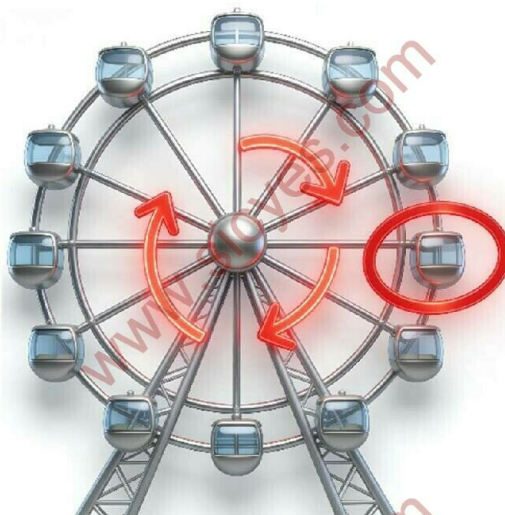
(Table tennis is very suitable for the characteristics of Asians and is the national sport of China.

Which of the following statements about table tennis is correct?)

- A. 乒乓球体积较小, 任何时候都可视为质点(Ping Pong is relatively small and can be viewed as a particle at any time)
 - B. 研究乒乓球的飞行轨迹时, 可将其视为质点(When studying the flight path of the ball, it can be viewed as a particle)
 - C. 乒乓球在空中飞行过程中的位移大小等于路程(The displacement of the ball during its flight in the air is equal to the distance traveled)
 - D. 研究乒乓球的旋转时, 必须将其视为质点(When studying the rotation of the ball, one must consider it as a particle)
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第 7 题(Question 7)

题目:摩天轮的半径为 R , 座舱随摩天轮转动一周, 其通过的路程和位移分别为(The radius of a Ferris wheel is R . The cabin rotates with the wheel once, and its distance and displacement are)

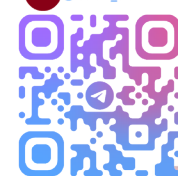


- A. 路程 $2\pi R$, 位移 $2R$ (distance $2\pi R$, displacement $2R$)
 - B. 路程 $2R$, 位移 $2\pi R$ (distance $2R$, displacement $2\pi R$)
 - C. 路程 $2R$, 位移 0 (distance $2R$, displacement 0)
 - D. 路程 $2\pi R$, 位移 0 (distance $2\pi R$, displacement 0)
-

第 8 题(Question 8)

题目:下列说法不正确的是(Which of the following statements is incorrect?)

- A. 在结冰的水平路面上撒一些细土, 人不易滑倒, 因为人与路面间的最大静摩擦力增大了
(Sprinkling some fine soil on a frozen horizontal road surface would make it less likely for people to slip on it, because the maximum static friction between the people and the road surface increases)
- B. 竖直握住空中的瓶子, 当手的握力增大时, 瓶子受到的摩擦力不变(Hold a bottle in the hand vertically in the air. When the grip of the hand is increased, the frictional force on

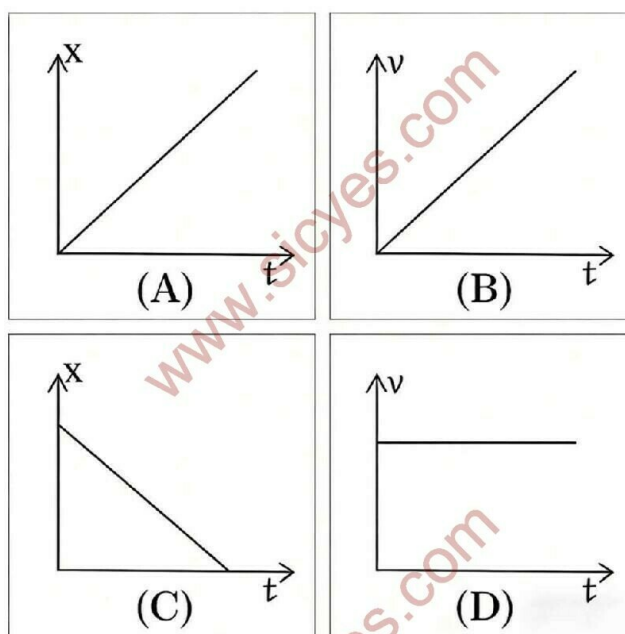


the bottle remains unchanged)

- C. 人行走时，受到的是静摩擦力(When walking, people are subjected to static friction)
- D. 人双手握住竖直固定的竹竿匀速上爬时，受到的摩擦力方向向下(When a person climbs a vertically fixed bamboo pole with both hands at a constant speed, the direction of the frictional force experienced is downward)

第 9 题(Question 9)

题目:下列哪个图像表示汽车做匀加速直线运动? (Which of the following graphs represents the motion of a car moving on a straight line with constant acceleration?)



第 10 题(Question 10)

题目:一个物体从 5 米高处自由下落，到达地面所需的时间为($g = 10, \text{m/s}^2$)(An object falls freely from a height of 5m. The time it takes to reach the ground is ($g = 10, \text{m/s}^2$))

- A. 4s(4 秒, 4 seconds)
- B. 1s(1 秒, 1 second)

C. 2s(2 秒, 2 seconds)

D. 3s(3 秒, 3 seconds)

第 11 题(Question 11)

题目:一个物体以 0.1 米为半径做匀速圆周运动, 周期为 5 秒, 该物体运动的线速度和角速度分别为(An object moves in a uniform circular motion with a radius of 0.1m and a period of 5s. The linear velocity and angular velocity of the object during motion are, respectively,)

A. $100\pi, \text{m/s}$, $0.4\pi, \text{rad/s}$ ($100\pi, \text{米/秒}$, $0.4\pi, \text{弧度/秒}$)

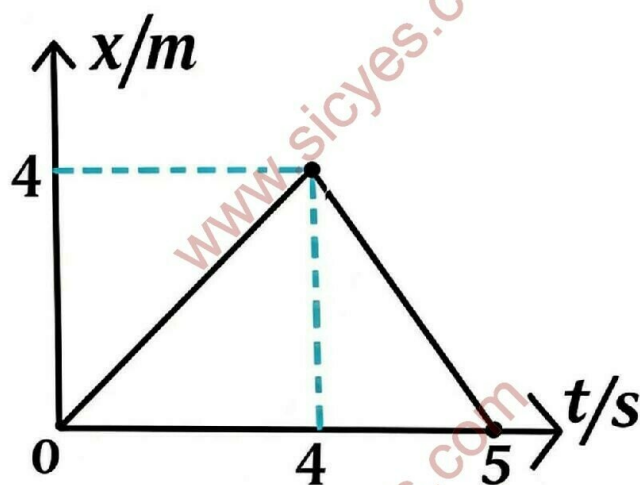
B. $0.4\pi, \text{m/s}$, $0.4\pi, \text{rad/s}$ ($0.4\pi, \text{米/秒}$, $0.4\pi, \text{弧度/秒}$)

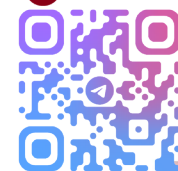
C. $0.5\pi, \text{m/s}$, $100\pi, \text{rad/s}$ ($0.5\pi, \text{米/秒}$, $100\pi, \text{弧度/秒}$)

D. $0.04\pi, \text{m/s}$, $0.4\pi, \text{rad/s}$ ($0.04\pi, \text{米/秒}$, $0.4\pi, \text{弧度/秒}$)

第 12 题(Question 12)

题目:某玩具车沿直线运动的位移(x)随时间(t)变化的图像如图所示, 下列说法正确的是(The graph of the displacement (x) of a toy car moving in a straight line changing with time (t) is shown in the figure. Which of the following statements is correct?)





- A. 0-4 秒内, 玩具车的位移为 8 米(Within 0-4 seconds, the displacement of the toy car's movement is 8m)
- B. 0-4 秒内, 玩具车的速度为 1.0 米/秒(Within 0-4 seconds, the speed of the toy car is 1.0m/s)
- C. 0-5 秒内, 玩具车的平均速度为 0.8 米/秒(Within 0-5 seconds, the average speed of the toy car is 0.8m/s)
- D. 4-5 秒内, 玩具车的速度为 0.8 米/秒(Within 4-5 seconds, the speed of the toy car is 0.8m/s)

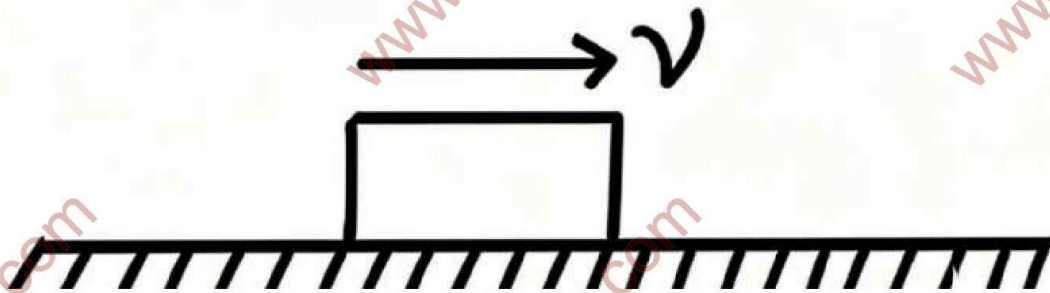
第 13 题(Question 13)

题目:当电梯以 $2, \text{m/s}^2$ 的加速度竖直向上加速运动时, 电梯内质量为 50 千克的人对地板的压力为($g = 10, \text{m/s}^2$)(When an elevator accelerates vertically upwards with an acceleration of $2, \text{m/s}^2$, the pressure exerted on the floor by a person with a mass of 50kg inside the elevator is ($g = 10, \text{m/s}^2$))

- A. 500N(500 牛, 500 newtons)
- B. 300N(300 牛, 300 newtons)
- C. 400N(400 牛, 400 newtons)
- D. 600N(600 牛, 600 newtons)

第 14 题(Question 14)

题目:如图所示, 一个质量为 1 千克的物体沿水平地面向右滑动, 已知物体受到的滑动摩擦力为 3 牛, $g = 10, \text{m/s}^2$, 则物体与水平地面间的动摩擦因数为(As shown in the figure, an object with a mass of 1kg slides to the right along a horizontal ground. Given that the sliding friction force acting on the object is 3N, and $g = 10, \text{m/s}^2$, the coefficient of kinetic friction between the object and the horizontal ground is)



A. 0.2(0.2)

B. 0.3(0.3)

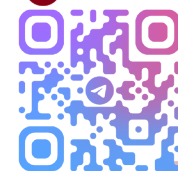
C. 1(1)

D. 0.4(0.4)

第 15 题(Question 15)

题目:在光滑水平面上, 对一个静止的质量为 1.0 千克的物体施加一个水平恒力 F, 测得其 1 秒内速度变化量的大小为 10 米/秒, 则力 F 的大小为(On a smooth horizontal surface, a constant force F is applied horizontally to a stationary object with a mass of 1.0kg, and the magnitude of its velocity change within 1 second is measured to be 10m/s. Therefore, the magnitude of the force F is)

A. 40N(40 牛, 40 newtons)



- B. 5N(5 牛, 5 newtons)
- C. 10N(10 牛, 10 newtons)
- D. 20N(20 牛, 20 newtons)

第 16 题(Question 16)

题目:光从空气射入某种介质时,入射角 $i = 60^\circ$, 折射角 $r = 30^\circ$, 若空气的折射率 $n = 1$, 则该介质的折射率为(When light enters a medium from air, with an angle of incidence $i = 60^\circ$, the angle of refraction is $r = 30^\circ$. If the refractive index of air is $n = 1$, the refractive index of the medium is)

- A. $\sqrt{3}$
- B. 3(3)
- C. 1.5(1.5)
- D. $\sqrt{5}$

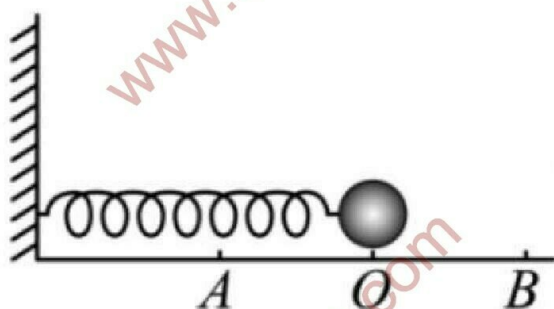
第 17 题(Question 17)

题目:一个小球以 $100\pi, \text{rad/s}$ 的角速度做匀速圆周运动, 小球到圆心的距离为 1 厘米, 则小球的向心加速度为(A small ball is in uniform circular motion with an angular velocity of $100\pi, \text{rad/s}$. The distance between the ball and the center of the circle is 1cm. The centripetal acceleration of the ball is)

- A. $50\pi^2, \text{m/s}^2$ ($50\pi^2, \text{米/秒}^2$)
- B. $100\pi^2, \text{m/s}^2$ ($100\pi^2, \text{米/秒}^2$)
- C. $10000\pi^2, \text{m/s}^2$ ($10000\pi^2, \text{米/秒}^2$)
- D. $\pi, \text{m/s}^2$ ($\pi, \text{米/秒}^2$)

第 18 题(Question 18)

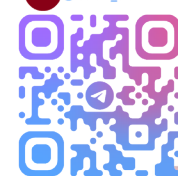
如图所示，弹簧振子在 A、B 之间做简谐运动，O 为平衡位置，测得 A、B 间距为 6cm，小球完成 30 次全振动所用时间为 60s，则(As shown in the figure, a spring oscillator performs simple harmonic motion between A and B, with O as the equilibrium position. The distance between A and B is measured to be 6 cm, and the time taken for the ball to complete 30 full oscillations is 60 s. Then,)



- A. 若从小球经过 O 点开始计时，3 秒内小球通过的路程为 24cm(If the timing starts when the ball passes point O, the distance traveled by the ball within 3 seconds is 24 cm)
- B. 小球完成一次全振动通过的路程为 12cm(The distance traveled by the ball in one complete oscillation is 12 cm)
- C. 该振子的周期为 2s，振幅为 6cm(The period of the oscillator is 2 s, and the amplitude is 6 cm)
- D. 该振子的频率为 2Hz(The frequency of the oscillator is 2 Hz)

第 19 题(Question 19)

题目:一个质量为 50 千克的物体以 10 米/秒的速度水平向右运动，其动量和动能分别为(An object with a mass of 50kg moves horizontally towards the right at a speed of 10m/s. The momentum and kinetic energy are, respectively,)



- A. 500,kg·m/s , 2500J(500,千克·米/秒 , 2500 焦耳, 2500 joules)
- B. 2500,kg·m/s , 500J(2500,千克·米/秒 , 500 焦耳, 500 joules)
- C. 500,kg·m/s , 5000J(500,千克·米/秒 , 5000 焦耳, 5000 joules)
- D. 5,kg·m/s , 500J(5,千克·米/秒 , 500 焦耳, 500 joules)

第 20 题(Question 20)

题目:一滴初速度为零的水滴从 5 米高的屋檐落下(不计空气阻力), 水滴落到地面时的速度为 ($g = 10, \text{m/s}^2$)(If a water droplet with zero initial velocity falls from a 5-meter high eaves (excluding air resistance), the speed at which the water droplet falls to the ground is ($g = 10, \text{m/s}^2$))

- A. 5m/s(5 米/秒, 5 meters per second)
- B. 10m/s(10 米/秒, 10 meters per second)
- C. 2m/s(2 米/秒, 2 meters per second)
- D. 20m/s(20 米/秒, 20 meters per second)

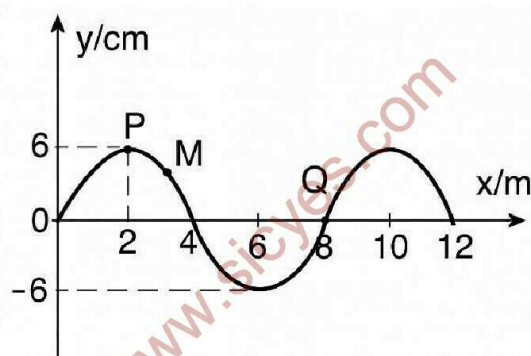
第 21 题(Question 21)

题目:一个木箱在与水平方向成 37° 角的拉力 $F = 50, \text{N}$ 作用下, 在水平面上移动了 10 米, 已知 $\cos 37^\circ \approx 0.8$, 则拉力 F 对木箱做的功为(A wooden box is pulled by a force $F = 50, \text{N}$ at an angle of 37° to the horizontal direction, which causes it to move a distance of 10m on a horizontal surface. Given that $\cos 37^\circ \approx 0.8$, the work done by the force F on the box is)

- A. 400J(400 焦耳, 400 joules)
- B. 300J(300 焦耳, 300 joules)
- C. 500J(500 焦耳, 500 joules)

D. 250J(250 焦耳, 250 joules)

第 22 题(Question 22)

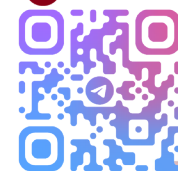


一列简谐横波沿 x 轴正方向传播, 波速为 10 米/秒, $t=0$ 时刻的波形如图所示, 下列说法正确的是(A simple harmonic transverse wave propagates along the positive direction of the x -axis, with a wave velocity of 10m/s. The waveform at $t=0$ is shown in the figure. Which of the following statements is correct?)

- A. 0~0.6 秒内, 质点 P 运动的路程为 18cm(Within 0~0.6s, the distance traveled by particle P is 18cm)
- B. $t=0.6$ s 时刻, 质点 P 相对平衡位置的位移是 6cm(At $t=0.6$ s, the displacement of particle P relative to the equilibrium position is 6cm)
- C. $t=1.2$ s 时刻, 质点 Q 加速度最大(At $t=1.2$ s, the acceleration of particle Q is maximum)
- D. $t=1.4$ s 时刻, 质点 M 沿 y 轴负方向运动(At $t=1.4$ s, particle M moves along the negative y -axis direction)
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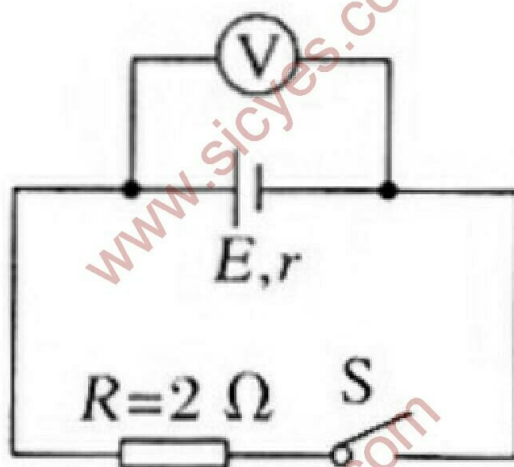
第 23 题(Question 23)

题目:两个电阻的阻值分别为 3 欧和 6 欧, 下列说法正确的是(Two electrical resistors have resistances 3Ω and 6Ω respectively. Which of the following statements is correct?)



- A. 串联时总电阻为 2 欧, 并联时总电阻为 2 欧(When connected in series, the total resistance is 2Ω ; when connected in parallel, the total resistance is 2Ω)
- B. 串联时总电阻为 9 欧, 并联时总电阻为 9 欧(When connected in series, the total resistance is 9Ω ; when connected in parallel, the total resistance is 9Ω)
- C. 串联时总电阻为 9 欧, 并联时总电阻为 2 欧(When connected in series, the total resistance is 9Ω ; when connected in parallel, the total resistance is 2Ω)
- D. 串联时总电阻为 2 欧, 并联时总电阻为 9 欧(When connected in series, the total resistance is 2Ω ; when connected in parallel, the total resistance is 9Ω)

第 24 题(Question 24)



题目: 如图所示的电路中, 电阻 $R = 2, \Omega$, 断开开关 S 后, 电压表的示数为 3 伏; 闭合开关 S 后, 电压表的示数为 2 伏, 则电源的内阻 r 为(In the circuit shown in the figure, the resistor has $R = 2, \Omega$. After disconnecting S, the voltmeter reading is 3V; after closing S, the voltmeter reading is 2V. So the internal resistance r of the power supply is)

- A. 3Ω (3 欧, 3 ohms)
- B. 2Ω (2 欧, 2 ohms)
- C. 4Ω (4 欧, 4 ohms)

D. 1Ω (1 欧, 1 ohm)

第 25 题(Question 25)

题目:一个气球内充有理想气体, 当气球上升时, 假设气体体积保持不变, 温度降低, 则在此过程中气球内的理想气体(A balloon is filled with ideal gas. When the balloon rises, assume that the volume of the gas remains constant, and the temperature decreases. In this process the ideal gas in the balloon would)

- A. 吸热(Absorb heat)
 - B. 压强增大(Increase pressure)
 - C. 内能减小(Decrease internal energy)
 - D. 对外做负功(Do negative work to outside)
-

第 26 题(Question 26)

题目:一个物体竖直上抛上升过程中, 重力对物体做功 -70 焦耳, 不计空气阻力, 下列说法正确的是(During the ascent of an object being thrown vertically upwards, gravity does -70J of work on the object. Ignoring air resistance, which of the following statements is correct?)

- A. 重力势能减少 70 焦耳(The gravitational potential energy decreases by 70J)
 - B. 重力势能增加 70 焦耳(The gravitational potential energy increases by 70J)
 - C. 机械能减少 70 焦耳(The mechanical energy decreases by 70J)
 - D. 重力势能保持不变(The gravitational potential energy remains constant)
-

第 27 题(Question 27)

题目:若通过一根导线的电流为 1.6 安, 则 0.5 秒内通过导线横截面的电荷量为(If the current



passing through a wire is 1.6A, the amount of charge passing through the cross-section of the wire within 0.5s is)

- A. 1.6C(1.6 库仑, 1.6 coulombs)
- B. 1.2C(1.2 库仑, 1.2 coulombs)
- C. 2.0C(2.0 库仑, 2.0 coulombs)
- D. 0.8C(0.8 库仑, 0.8 coulombs)

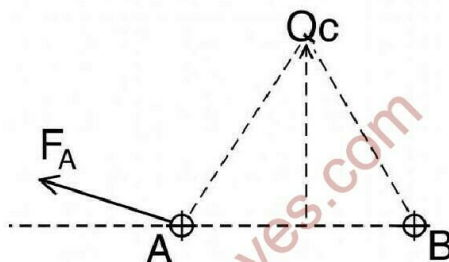
第 28 题(Question 28)

题目:一定质量的理想气体, 温度保持不变, 压强减小, 则其体积会(A certain mass of ideal gas, with its temperature kept constant and pressure reduced, its volume would)

- A. 选项均有可能(All options above are possible)
- B. 增大(Increase)
- C. 减小(Decrease)
- D. 保持不变(Remain unchanged)

第 29 题(Question 29)

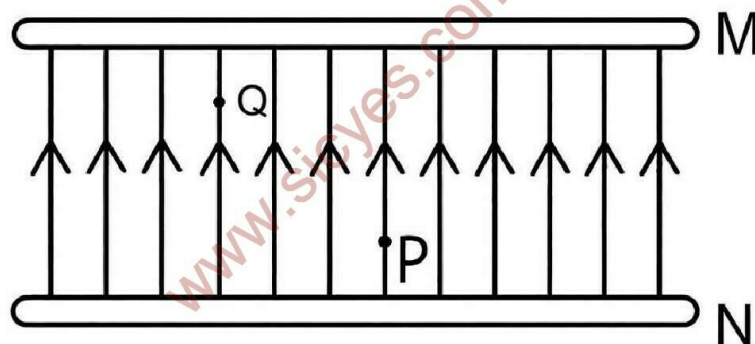
题目:三个点电荷 A、B、C 位于等边三角形的三个顶点, A、B 均带正电, 电荷 B、C 对 A 的静电力的合力 $\vec{F}_A$ 如图所示, 已知 $\vec{F}_A$ 与 BA 的延长线的夹角小于 60° , 则关于电荷 C 的正确判断是(Three point charges A, B, and C are located at the vertices of an equilateral triangle, with



both A and B carrying positive charges. The resultant forces of the electrostatic forces of the two charges B and C acting on A is shown as $\vec{F}_A$ in the figure. Given that the angle between $\vec{F}_A$ and the extension line of BA is less than 60° , what is the correct judgment about the charge at C?)

- A. 带正电, 电荷量小于 B (Positively charged, charge amount less than B)
- B. 带负电, 电荷量大于 B (Negatively charged, charge amount greater than B)
- C. 带正电, 电荷量大于 B (Positively charged, charge amount greater than B)
- D. 带负电, 电荷量小于 B (Negatively charged, charge amount less than B)

第 30 题 (Question 30)



题目: 如图所示, 两个带等量异种电荷的平行金属板 M、N 水平放置, 两板间的电场线是间隔相等的平行线, P、Q 是电场中的两个点, 下列说法正确的是 (As shown in the figure, two parallel metal plates M and N with equal and opposite charges are placed horizontally. The electric field lines between them are parallel lines with equal intervals, and P and Q are two points in the electric field. Which of the following statements is correct?)

- A. M 带正电, N 带负电 (M has positive charge, N has negative charge)
- B. P、Q 两点的场强大小相等 (P, Q have equal field strength)
- C. P、Q 两点的电势和场强均不同 (Both the electrical potential and field strength at P, Q are different)



different)

D. P、Q 两点的电势相等(P, Q have equal electrical potential)

第 31 题(Question 31)

题目:一根长度 $L = 0.5, \text{m}$ 、通有电流 $I = 2, \text{A}$ 的直导线放在磁感应强度 $B = 0.4, \text{T}$ 的匀强磁场中, 磁场方向与导线中电流方向的夹角为 30° , 则导线受到的安培力大小为(A straight wire with length $L = 0.5, \text{m}$ and a current of $I = 2, \text{A}$ is placed in a uniform magnetic field with a magnetic induction intensity of $B = 0.4, \text{T}$. The angle between the direction of the magnetic field and the direction of the current in the wire is 30 degrees. The ampere force acting on the wire is)

- A. 0.4N (0.4 牛, 0.4 newtons)
- B. 0N (0 牛, 0 newtons)
- C. 0.2N (0.2 牛, 0.2 newtons)
- D. $\frac{\sqrt{3}}{5}, \text{N}$ ($\frac{\sqrt{3}}{5}$, 牛)

第 32 题(Question 32)

题目:在磁感应强度 $B = 0.8, \text{T}$ 的匀强磁场中, 一个矩形线圈的面积 $S = 0.5, \text{m}^2$, 线圈平面的法线方向与磁场方向的夹角为 60° , 则穿过线圈的磁通量大小为(In a uniform magnetic field with a magnetic induction intensity of $B = 0.8, \text{T}$, a rectangular coil has area $S = 0.5, \text{m}^2$, and the angle between the normal direction of the coil plane and the direction of the magnetic field is 60 degrees. Therefore, the magnitude of the magnetic flux passing through the coil is)

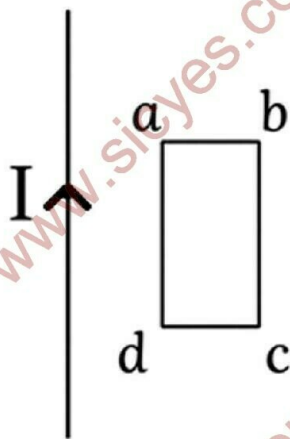
- A. 0.2Wb (0.2 韦伯, 0.2 webers)
- B. 0.8Wb (0.8 韦伯, 0.8 webers)

C. 0.4Wb(0.4 韦伯, 0.4 webers)

D. 0.6Wb(0.6 韦伯, 0.6 webers)

第 33 题(Question 33)

题目:如图所示, 在一根通有图示方向电流 I 的长直导线右侧, 固定一个矩形金属线框 $abcd$, 其边 bc 与导线平行。通过调节电流 I , 使空间各点的磁感应强度随时间均匀增大, 则(As shown in the figure, on the right side of a long straight wire with a current I flowing in the direction indicated in the diagram, a rectangular metal wire frame $abcd$ is fixed, with its side ad parallel to the wire. By adjusting the current I , the magnetic induction intensity at each point in space increases uniformly over time. Then,)



- A. 线框中感应电流的方向为 $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$ (The induced current in the frame has direction $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$)
- B. 线框中的感应电流逐渐增大(The induced current in the frame gradually increases)
- C. 线框受到水平向右的安培力(The frame experiences an ampere force horizontally towards the right)
- D. 边 bc 受到的安培力大小不变(The ampere force acting on the edge cd is constant)
-



第 34 题(Question 34)

题目:如图所示, 空间存在一个电场, a、b 为电场中的两个点, 下列说法正确的是(As shown in the figure, there is an electric field in space with two points a and b. Which of the following

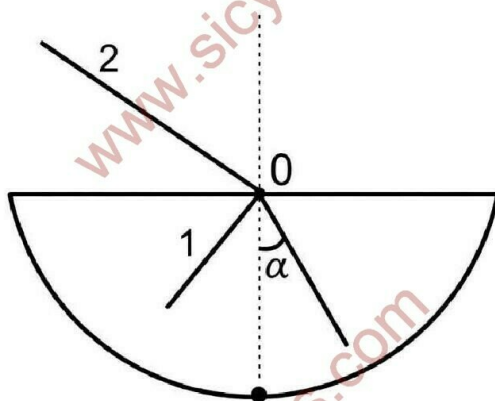


statements is correct?)

- A. b 点的电场强度大于 a 点(The electric field strength at point b is greater than that at point a)
 - B. a 点的电场强度大于 b 点(The electric field strength at point a is greater than that at point b)
 - C. 该电场为匀强电场(This electric field is a uniform electric field)
 - D. a、b 两点的电场强度大小相等(The electric field strength at points a and b is the same)
-

第 35 题(Question 35)

题目:如图所示, 一个玻璃圆柱的横截面为半圆形, 一束单色细光沿圆柱射向半圆的圆心 O , 入射角 $\alpha = 30^\circ$, 产生的反射光 1 和折射光 2 恰好垂直, 则玻璃的折射率为(As shown in the figure, the cross-section of a glass cylinder is semi-circular. A thin monochromatic beam travels through the cylinder towards the center of the semicircle O at an incident angle of $\alpha = 30^\circ$.

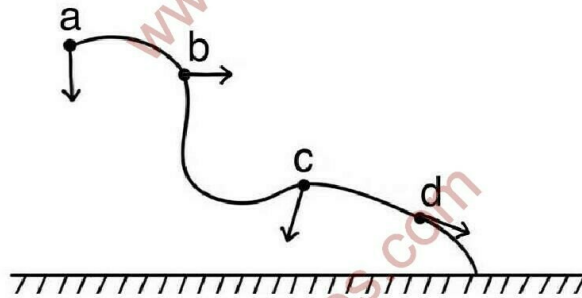


The reflected beam 1 and refracted beam 2 produced are exactly perpendicular to each other. Therefore, the refractive index of the glass is)

- A. $\sqrt{3}$
- B. $\sqrt{2}$
- C. $\frac{1}{\sqrt{3}}$
- D. $\frac{\sqrt{3}}{2}$



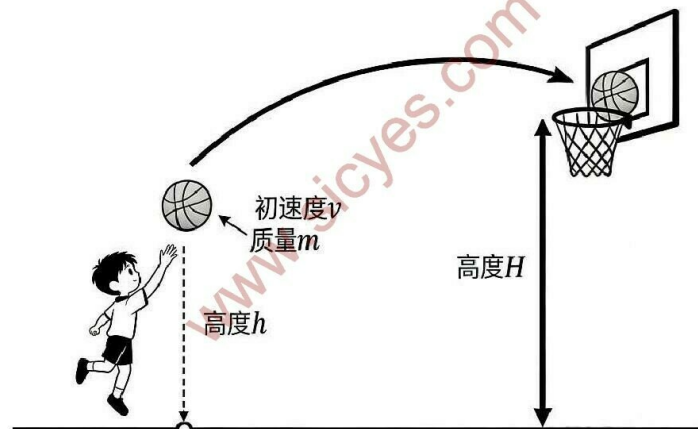
第 36 题(Question 36)



题目:如图所示, 一片树叶从空中下落的轨迹如图, 图中标记了树叶在 a、b、c、d 四个位置的速度方向, 其中正确的是(The trajectory of a leaf falling from the air is shown in the figure, the velocity directions of the leaf at positions a, b, c, and d are indicated. The correct direction is)

- A. a
- B. b
- C. c
- D. d

第 37 题(Question 37)



题目:如图所示, 一名学生将质量为 m 的篮球从高度 h 处以初速度 v 抛出, 篮球恰好进入高度为 H 的篮筐。以地面为零势能面, 不计空气阻力, 重力加速度大小为 g , 则篮球进入篮筐时的

机械能为(As shown in the figure, a student throws a basketball with a mass of m from a height of h at an initial velocity of v , and the basketball just enters the basket with a height of H above the ground. Taking the ground as the zero potential energy surface, ignoring air resistance, and let the magnitude of gravitational acceleration be g , the mechanical energy of the basketball entering the basket is)

- A. $\frac{1}{2}mv^2 + mgH$
 - B. $\frac{1}{2}mv^2 + mg(H - h)$
 - C. $\frac{1}{2}mv^2 + mgh$
 - D. $\frac{1}{2}mv^2 + mgH + mg(H - h)$
-

第 38 题(Question 38)

题目:一辆质量为 2 吨的汽车, 发动机功率为 30 千瓦, 在水平地面上以 72 千米/时的速度匀速行驶, 则汽车在该路面行驶时受到的阻力为(A car with a mass of 2 tons and an engine with a power of 30kW travels at a constant velocity of 72km/h on a level ground. The resistance experienced by the car while driving on that road is)

- A. $5 \times 10^3, \text{N}$ ($5 \times 10^3, \text{牛}$)
 - B. $6 \times 10^3, \text{N}$ ($6 \times 10^3, \text{牛}$)
 - C. $1.5 \times 10^3, \text{N}$ ($1.5 \times 10^3, \text{牛}$)
 - D. $2 \times 10^3, \text{N}$ ($2 \times 10^3, \text{牛}$)
-

第 39 题(Question 39)

题目:一个小球从某一高度以初速度 v_0 水平抛出, 落到水平地面上。随后在同一高度以初速度 $2v_0$ 再次水平抛出, 落到同一地面, 不计空气阻力, 则第二次抛出时(A ball is thrown horizontally

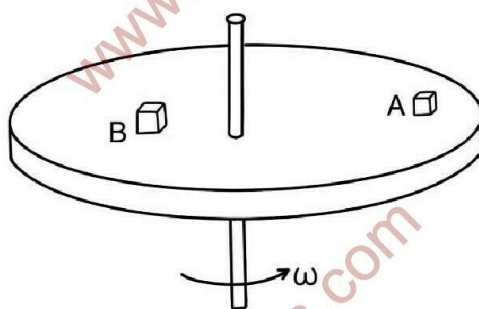


from a certain height at an initial velocity of v_0 and it lands on a level ground. Then, the ball is thrown again at the same height horizontally at an initial velocity of $2v_0$ and it lands on the same ground. Ignoring air resistance, during the second throw of the ball,)

- A. 水平位移变为原来的 2 倍(The horizontal displacement becomes twice of the original)
- B. 下落时间变为原来的 2 倍(The time taken to fall becomes twice of the original)
- C. 水平位移变为原来的 1/2(The horizontal displacement becomes half of the original)
- D. 下落时间变为原来的 1/2(The time taken to fall becomes half of the original)

第 40 题(Question 40)

题目:如图所示, 两个质量相等的物体 A、B 放在绕竖直轴匀速转动的水平圆盘上, 且两物体相对圆盘静止, 则(As shown in the figure, if two objects A and B with equal mass are placed on a horizontal disk rotating uniformly around the vertical axis, and the two objects remain stationary



relative to the disk, then)

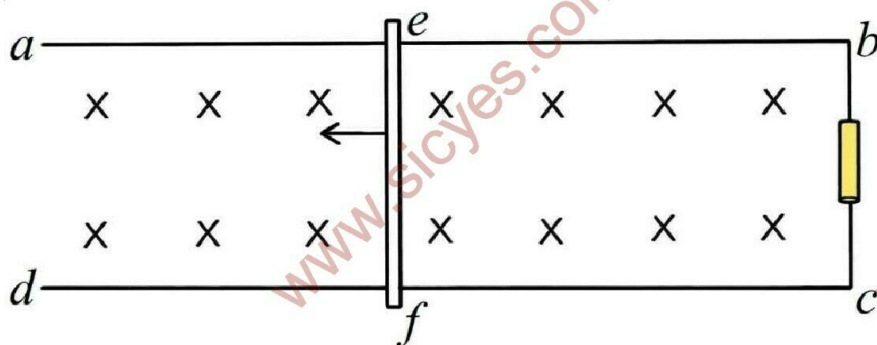
- A. A 的角速度与 B 的角速度相等(The angular velocity of A is the same as B)
- B. A 的线速度与 B 的线速度相等(The linear velocity of A is the same as B)
- C. A 的角速度大于 B 的角速度(The angular velocity of A is greater than that of B)
- D. A 的角速度小于 B 的角速度(The angular velocity of A is smaller than that of B)

第 41 题(Question 41)

题目:一个质量为 m 的物体从静止开始自由下落, 不计空气阻力, 经过时间 t 时重力对物体做功的功率为(An object with mass m falls freely from rest. Ignoring air resistance, the power delivered by gravity on the object over time t is)

- A. $\frac{1}{2}mg^2t$
- B. mgt
- C. mg^2t
- D. $\frac{1}{4}mgt$

第 42 题(Question 42)



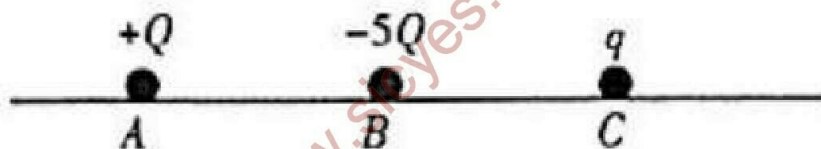
如图所示, 连接负载的固定 U 型金属导轨 abcd, 放置在与导轨平面垂直的匀强磁场中。导轨上有一根可自由滑动的金属棒 ef, 在金属棒向左平移的过程中。下列说法正确的是(As shown in the figure, a fixed U-shaped metallic rail abcd connected with a load is placed in a uniform magnetic field perpendicular to the plane of the rail. There is a freely slidable metal rod ef on the rail. During the leftward translation of the metal rod, which of the following statements is correct?)

- A. 穿过 ebcf 回路的磁通量不变(The magnetic flux passing through the ebcf loop remains unchanged)



- B. 穿过 ebcf 回路的磁通量变小(The magnetic flux passing through the ebcf loop decreases)
- C. ae 段有电流通过(There is a current flowing through segment ae)
- D. bc 段有电流通过(There is a current flowing through segment bc)

第 43 题(Question 43)



题目:如图所示, 两个相同的金属球(均可视为点电荷)固定在真空中的 A、B 两点, 电荷量分别为 $+Q$ 和 $-5Q$, 在 A、B 的延长线上有一点 C, 固定一个点电荷 q , 该电荷受到的静电力大小为 F_1 , 已知 $AB = BC$ 。若将两个带电金属球接触后再放回原处, 则点电荷 q 受到的静电力大小变为 F_2 , 则 $\frac{F_1}{F_2}$ 为(As shown in the figure, two identical metal balls (both of which can be considered as point charges) are fixed at points A and B in vacuum, with charges of $+Q$ and $-5Q$, respectively. A point charge q is fixed at point C on the line extending from A to B. The magnitude of the electrostatic force acting on this charge is F_1 . It is given that $AB = BC$. If the two charged metal balls are brought into contact and then returned to their original positions, the magnitude of the electrostatic force acting on charge q becomes F_2 . Thus, $\frac{F_1}{F_2}$ is)

- A. $\frac{21}{10}$
- B. $\frac{21}{16}$
- C. $\frac{19}{10}$
- D. $\frac{19}{16}$

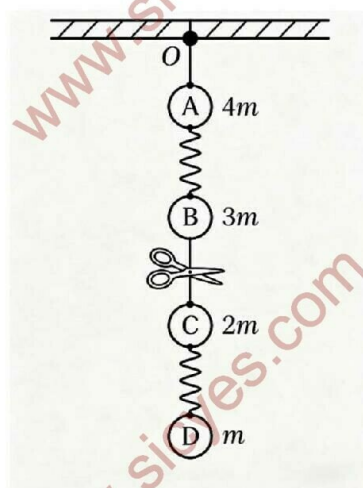
第 44 题(Question 44)

题目:一个质量 $m = 0.6, \text{kg}$ 的篮球以 $v_1 = 2, \text{m/s}$ 的速度水平向左运动, 撞击坚硬的墙壁后,

以 $v_2 = 1, \text{m/s}$ 的速度水平向右反弹，若球与墙壁的接触时间为 0.1 秒，则墙壁对篮球的平均水平作用力大小为(不计空气阻力， $g = 10, \text{m/s}^2$)(A basketball with a mass of $m = 0.6, \text{kg}$ moves horizontally to the left at a speed of $v_1 = 2, \text{m/s}$. When it hits a hard wall, it bounces back horizontally to the right at a speed of $v_2 = 1, \text{m/s}$. If the ball is in contact with the wall for 0.1 seconds, the average horizontal force exerted on the basketball by the wall is (Neglect air resistance, $g = 10, \text{m/s}^2$))

- A. 1.8N (1.8 牛, 1.8 newtons)
- B. 12N (12 牛, 12 newtons)
- C. 18N (18 牛, 18 newtons)
- D. 6N (6 牛, 6 newtons)

第 45 题(Question 45)



题目:如图所示，四个质量分别为 $4m$ 、 $3m$ 、 $2m$ 、 m 的小球 A、B、C、D 通过细导线或轻弹簧相互连接，悬挂在 O 点，处于静止状态，重力加速度为 g 。若剪断 B、C 之间的细导线，剪断瞬间 B、C 的加速度分别为(As shown in the figure, four small balls A, B, C, and D with masses of $4m$, $3m$, $2m$, and m , respectively, are connected to each other through thin wires or light springs, suspended at point O, in a stationary state with a gravitational acceleration of g . If the

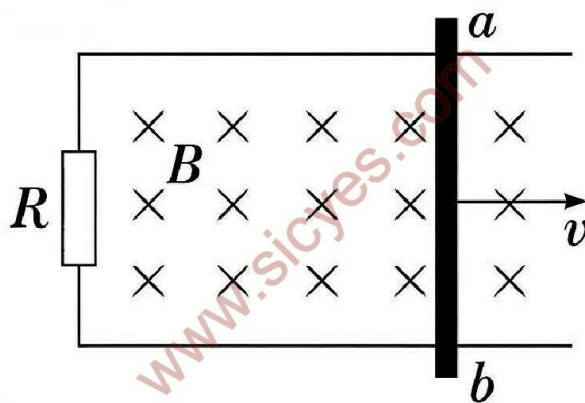


thin wire between B and C is cut, the accelerations of B and C at the instant the wire is cut are, respectively:)

- A. $2g, 1.5g$
- B. $g, 1.5g$
- C. $g, 0.5g$
- D. $2g, 0.5g$

第 46 题(Question 46)

题目:如图所示, 水平放置的平行金属导轨两端连接一个电阻 R , 导线 ab 可在框架上无摩擦滑动, 一个匀强磁场垂直穿过框架平面。当 ab 向右匀速运动时, 下列说法正确的是(As shown in the figure, the two ends of parallel metal rails placed horizontally are connected with a resistor R . The wire ab can slide on the frame without friction, and a uniform magnetic field passes vertically through the frame plane. When ab moves uniformly to the right, which of the following statements is correct?)

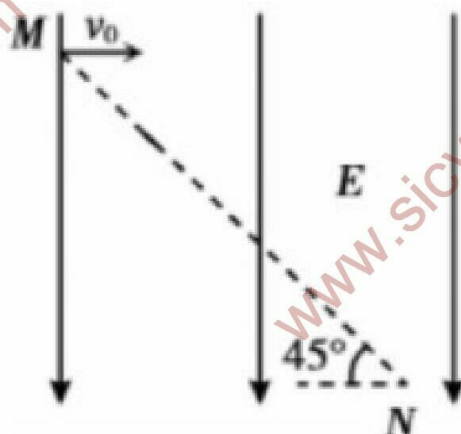


- A. 只要导线 ab 达到一定速度, 撤去外力, 导线 ab 也能在框架上保持匀速运动(As long as the wire ab reaches a certain velocity and the external force is removed, the wire ab can also maintain a uniform motion on the frame)

- B. 导线 ab 不仅受到拉力, 还受到磁场力(Wire ab is not only subjected to tension force, but also to magnetic field force)
- C. 导线 ab 的运动速度恒定, 若增大电阻 R 的阻值, 拉动导线 ab 的力可以增大(The movement velocity of wire ab is constant. If the resistance value R is increased, the force pulling wire ab can be increased)
- D. 导线 ab 的运动速度越高, 所需的拉力越小(The higher the velocity of wire ab's movement, the smaller the required pulling force)

第 47 题(Question 47)

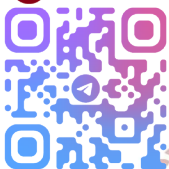
题目:如图所示, 一个质量为 m 、电荷量 $q > 0$ 的粒子, 以初速度 v_0 水平射入竖直向下的匀强电场中, 电场强度大小为 E 。粒子从直线 MN 上的 P 点射出, 已知直线 MN 与水平方向成 45° 角, 不计重力, 当粒子回到直线 MN 时



As shown in the figure, a particle of mass m and charge $q > 0$ is projected horizontally into a uniform electric field of magnitude E directed vertically downward. It is projected from point P on the line MN with an initial velocity v_0 . Given that line MN makes an angle of 45° with the horizontal, and assuming that gravity is negligible, when the particle returns to the line MN:

and assuming that gravity is negligible, when the particle returns to the line MN:)

- A. 速度与竖直方向的夹角为 30° (The angle that the velocity makes with the vertical direction is 30°)
- B. 到 P 点的距离为 $\frac{2\sqrt{2}mv_0^2}{qE}$ (The distance from P is $\frac{2\sqrt{2}mv_0^2}{qE}$)
- C. 所用时间为 $\frac{mv_0}{qE}$ (The time it takes is $\frac{mv_0}{qE}$)
- D. 速度大小为 $3v_0$ (The velocity is $3v_0$)



第 48 题(Question 48)

题目:高速行驶时保持安全车距至关重要。某小型汽车的最大加速度大小为 $6, \text{m/s}^2$, 一名司机在高速公路上以 108 千米/时的速度行驶, 若司机的反应时间为 1.5 秒, 则与前车的最小安全距离至少为(It is important to maintain a safe driving distance at high speeds. The maximum deceleration of a small car is $6, \text{m/s}^2$. A driver is driving a car on a highway at a speed of 108km/h. If the driver's reaction time is 1.5 seconds, the minimum safe distance from the car ahead should be at least)

- A. 90m(90 米, 90 meters)
- B. 60m(60 米, 60 meters)
- C. 120m(120 米, 120 meters)
- D. 150m(150 米, 150 meters)